

Nitish Rathore

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EDUCATION

University of Massachusetts Amherst

Sep. 2026 – May 2028 (Expected)

Master of Science in Computer Science

Amherst, MA

Coursework: Machine Learning, Trustworthy & Responsible AI, Theory & Practice of Software Engineering

PES University

Dec. 2020 – May 2024

Bachelor of Technology in Computer Science and Engineering

GPA: 8.66/10

Coursework: Big Data, Database Systems, Data Structures & Algorithms, Operating Systems, Cloud Computing

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Java, C#, JavaScript, TypeScript

Deep Learning & CV: PyTorch, TensorFlow, OpenCV, YOLO, MediaPipe, Faster R-CNN, ResNet, SAM, ONNX

ML & Data Science: scikit-learn (Random Forest, DBSCAN, K-Means), SciPy, NumPy, Pandas, FastDTW, Time-Series

LLMs & GenAI: Gemini (Vertex AI), OpenAI API, Multimodal Prompting, Prompt Engineering

ML Systems & Data: Ray, Ray Serve, GPU Inference, RabbitMQ, Kafka, ZooKeeper, ClickHouse, MongoDB, ffmpeg

Cloud & DevOps: AWS (S3, EC2, ECS, Lambda, SQS), Docker (GPU), Kubernetes, GitHub Actions, CI/CD, Linux, Git, Pytest

EXPERIENCE

Pronisi, Inc. | *Software Engineer, ML*

Jun. 2024 – Feb. 2025

- Led computer vision for **SlamDunk.AI**, an AI coaching app for tennis, cricket, basketball, golf, yoga, and gym.
- Shipped YOLOv7/v8 ball, racket, and golf-club detectors with FP16 traced PyTorch and ONNX Runtime (CUDA/CPU) at 1280 px input, cutting inference time by **40%**.
- Raised ball-tracking reliability by clustering detections with **DBSCAN** to reject false positives, interpolating gaps, and splitting multi-stroke clips with **K-Means** over detection gaps.
- Turned 30-keypoint 3D pose (MediaPipe, with jitter detection) into form scores: rule-based swing-phase detection, **FastDTW** frame alignment for dance, and SciPy peak detection for rep counting.
- Contributed to the hybrid **recommendation engine**: severity-scored tennis faults with linear-regression trend analysis, and a **Random Forest** flagging wrong squat, push-up, and crunch reps from proximity/stability scores.
- Used **multimodal LLMs** for coaching: Gemini 1.5 (Vertex AI) compares user and pro video, frames, and joint angles per rep, with parallel calls and exponential-backoff retries; GPT personalizes tennis feedback.
- Ran pose, ball, racket, and baseline models as concurrent GPU Ray actors; added **24 pytest cases** across 20 modules.

Pronisi, Inc. | *Machine Learning Intern*

Jun. 2023 – May 2024

- Built real-time pose estimation and keypoint-based equipment detection with MediaPipe, YOLO, Faster R-CNN, ResNet, and SAM for tennis, cricket, basketball, and golf.
- Reached **80%+ accuracy** tracking fast-moving balls and equipment with YOLOv7/8 and trajectory smoothing.
- Managed the ML data pipeline end to end: video collection, cleaning, annotation, and preprocessing.
- Built anchor-frame video synchronization and split-screen comparison of user and professional motion.

KangaSys | *Software Engineer, Data & Analytics Platform*

Mar. 2025 – Jun. 2026

- Built a multi-tenant analytics platform on ClickHouse and AWS ingesting **10M+ daily sensor events** from Honeywell systems, with **100–500 ms** queries over 1M+ record datasets.
- Built Next.js dashboards (30+ configurable widgets) and serverless PDF/CSV reports (Lambda + SQS) with tenant RBAC.

IIIT Bangalore | *Research Intern*

Jun. 2022 – Aug. 2022

- Built CNN classifiers for multispectral satellite land-use/land-cover mapping, with noise-reduction and cloud-masking preprocessing and statistical band selection to cut compute at similar accuracy.
- Evaluated spatial accuracy with GRASS GIS, GDAL, and Rasterio, contributing to roughly 90% accuracy.

PROJECTS & PUBLICATIONS

Urban Growth Modeling: Cellular Automata + GA | *Python, Keras, GIS*

IEEE BDAI 2023

- Predicted land-use/land-cover change with Cellular Automata whose parameters were optimized by a Genetic Algorithm, reaching **SSIM 0.84** against 2017–2022 satellite maps; awarded **Best Presenter**.
- Produced the land-cover maps with a Keras neural-network classifier on Landsat-5 imagery (built-up, vegetation, barren, water), reaching **F1 0.93** and roughly 90% area-wise accuracy on the held-out year.
- Compared Google Maps, OSRM, and OSMnx routes to find congestion-prone corridors; proposed transit and road upgrades from spatial-demand and hydraulic (Hazen-Williams) models.

Kafka from Scratch | *Python, ZooKeeper, RabbitMQ, Redis*

github-link

- Built a Kafka-style system (sockets and RabbitMQ) with ZooKeeper coordination, key-hash partitioning, 3-way replication, leader election, heartbeat monitoring, and automatic failover.